

Contact

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Academic Positions

Perimeter Institute for Theoretical Physics – Waterloo, Canada Sept 2023 - Present
Postdoctoral Fellow
Mentor: Roger Melko

Harvard John A. Paulson SEAS – Boston, USA May 2023 - Aug 2023
Postdoctoral Fellow (Applied Math)
Supervisor: Cengiz Pehlevan

Education

Northeastern University – Boston, USA 2017-2023
Ph.D. (Physics)
Advisor: James Halverson

The NSF AI Institute for Artificial Intelligence and Fundamental Interactions – Boston
Junior Investigator 2020-2023

Indian Institute of Technology Bombay – Mumbai, India 2012-2017
Integrated Bachelor and Master of Technology (Engineering Physics) with Honors (Physics)
Advisor: Urjit Yajnik

Research Interests

Theoretical physics + applied mathematics for Artificial Intelligence (AI).
Bridging theory-practice gap in AI interpretability, safety and robustness.
AI for quantum field theory and quantum.

Preprints & Publications

L. Greenspan, D. Berman, A. Brill, R. Jefferson, A. Kolchinsky, J. Lin, A. Mack, **A. Maiti**, F. E. Rosas, A. Stapleton, L. Teixeira, D. Vaintrob, “Towards Worst-Case Guarantees with Scale-Aware Interpretability”, [[arXiv:2602.05184](#)].

S. Frank, J. Halverson, **A. Maiti**, F. Ruehle, “Fermions and Supersymmetry in Neural Network Field Theories”, [[arXiv:2511.16741](#)].

J. Q. Toledo-Marin, **A. Maiti**, G. C. Fox, R. G. Melko, “Exploring the Energy Landscape of RBMs: Reciprocal Space Insights into Bosons, Hierarchical Learning and Symmetry Breaking”, *Mach. Learn.: Sci. Technol.* **6** 035030 (2025), [[arXiv:2503.21536](#)].

J. N. Howard, M. S. Klinger, **A. Maiti**, A. G. Stapleton, “Bayesian RG Flow in Neural Network Field Theories”, *SciPost Physics Core*, **8**(1), 027, [[arXiv:2405.17538](#)].

Y. M. Lu, M. I. Letey, J. A. Zavatone-Veth, **A. Maiti**, C. Pehlevan, “Asymptotic theory of in-context learning by linear attention”, *Proceedings of the National Academy of Sciences* **122** (28), e2502599122, [[arXiv:2405.11751](#)].

Y. M. Lu, M. I. Letey, J. A. Zavatone-Veth, **A. Maiti**, C. Pehlevan, “In-Context Learning by Linear Attention: Exact Asymptotics and Experiments”, [[NeurIPS 2024 Workshop on Mathematics of Modern Machine Learning](#)].

J. N. Howard, R. Jefferson, **A. Maiti**, Z. Ringel, “Wilsonian Renormalization of Neural Network Gaussian Processes”, *Machine Learning: Science and Technology* **6** (2), 025038 (2025), [[arXiv:2405.06008](#)].

M. Demirtas, J. Halverson, **A. Maiti**, M. D. Schwartz, K. Stoner, “Neural Network Field Theories: Non-Gaussianity, Actions, and Locality”, *Mach. Learn. Sci. Tech.* **5**, 015002 (2024), [[arXiv:2307.03223](#)].

A. Maiti, K. Stoner, J. Halverson, “Symmetry-via-Duality: Invariant Neural Network Densities from Parameter-Space Correlators”, *MACHINE LEARNING: IN PURE MATHEMATICS AND THEORETICAL PHYSICS*, 2023, **293-330**, [[arXiv:2106.00694v1](#)].

J. Halverson, C. Long, **A. Maiti**, B. Nelson, G. Salinas, “Gravitational waves from dark Yang-Mills sectors”, *JHEP* **05** (2021), 154, [arXiv:2012.04071].

J. Halverson, **A. Maiti**, K. Stoner, “Neural Networks and Quantum Field Theory”, *Mach. Learn. Sci. Tech.* **2** (2021) no. 3, 035002, [arXiv:2008.08601].

Awards & Honors

UC Riverside Chancellor’s Postdoctoral Fellowship: Jul 1, 2023 to Jul 1, 2024 (Declined).

Dean’s Graduate Student Excellence Award in Research: Northeastern University College of Science (Spring 2021).

Lawrence Award for Graduate Academic Excellence: Northeastern University Dept. of Physics (Spring 2018).

Travel Grants: “Theoretical Physics for Machine Learning” Workshop by Aspen Center for Physics (Feb 2023); “New Frontiers in Machine Learning and Quantum” Workshop by Perimeter Institute (Nov 2022); The NSF IAIFI (Feb 2023); Northeastern University Dept. of Physics (Summer 2022); Northeastern University PhD Network (Summer 2022).

Indian Academy of Sciences Summer Research Fellowship: Indian Academy of Sciences, (Summer 2014).

Seminars, Talks, Colloquia

Invited Workshop, Conference Talks, Panel Discussions

Renormalization for AI Safety Workshop, Principles of Intelligent Behavior in Biological and Social Systems, Microsoft Conference Center New England Oct 2025

Probing the Frontiers of Nuclear Physics with AI at the EIC, Stony Brook University CFNS Mar 2025

No Stone Unturned 2025, University of Utah, USA Mar 2025

Panel Discussion: The Intersection of AI, Physics, and Tech Innovation, Perimeter Institute in partnership with Volta, Halifax, Nova Scotia, Canada Feb 2025

String Data 2024, Kyoto University, Japan Dec 2024

Panel Discussion: She Talks Tech, The Accelerator Center, Kitchener, Canada July 2024

Machine Learning and the Renormalization Group, ECT*, Italy May 2024

Future Horizons: Bridging AI, Quantum and New Materials, IVADO, Perimeter Institute, Institut Courtois and Mila Quebec, Canada May 2024

Fields, Strings, and Deep Learning, Aspen Center for Physics Jan 2024

(Non-speaking) Program Invitation: Deep Learning from the Perspective of Physics and Neuroscience, Kavli Institute for Theoretical Physics, UCSB Nov 2023

AI and Quantum Information for Particle Physics, KAIST and IBS Center for Theoretical Physics of the Universe Nov 2023

Probing the Frontiers of Nuclear Physics with AI at the EIC, Stony Brook University CFNS Sept 2023

Machine Learning for Lattice Field Theory and Beyond, ECT*, Italy Jun 2023

New Frontiers in Machine Learning and Quantum, Perimeter Institute Nov 2022

Short Talks: A Deep-Learning Era of Particle Theory, Mainz Institute for Theoretical Physics, Johannes Gutenberg University June 2022

String Data 2021, University of Witwatersrand & University of Cape Town Dec 2021

Invited Colloquia and Seminars

High Energy Theory Seminar, Carnegie Mellon University May 2026

Joint Dept. of Theoretical Physics and School of Technology and Computer Science Colloquium, Tata Institute of Fundamental Research (TIFR), Mumbai, India Mar 2026

Bay Area Particle Theory Seminar, Theoretical Physics Group, Lawrence Berkeley National Laboratory Dec 2025

High-TheQ group seminar series , Ghent University, Belgium	Jul 2025
CHEP Seminar , Indian Institute of Science, Bangalore	Jan 2025
Strings Seminar , International Center for Theoretical Sciences, Bangalore	Jan 2025
High Energy Physics Seminar , Indian Institute of Technology Kanpur	Jan 2025
High Energy Physics Seminar , Indian Institute of Technology Kanpur	June 2024
Stanford Linear Accelerator Center (SLAC) AI Seminar , Stanford University	Feb 2024
ORIGINS Data Science Lab Seminar , TU Munich	Jul 2023
Center for Theoretical Physics Seminar , Seoul National University	Mar 2023
AIC Seminar , Université Paris-Saclay, CEA-LIST	Jan 2023
UCI Physics Astro/Particle-ML Seminar Series , UC Irvine	Oct 2022
UCSB Joint HEX-HET Seminar Series , UC Santa Barbara	Oct 2022
HEP Seminar , UC Riverside	Oct 2022
Majorana-Raychaudhuri Seminar Series , INFN & University Salerno, Italy & PAMU, Indian Statistical Institute, Kolkata, India	Sept 2022
Computational Algebra Seminar Series , University of Nottingham, UK	Sept 2022
Pehlevan Research Group Journal Club , Harvard University	Aug 2022
QFT Research Seminar , Institute for Theoretical Physics - Münster (WWU)	May 2021
Joint High Energy Theory and Machine Learning Seminar , Heidelberg University, LMU Munich and Northeastern University	May 2021
Journal Club , The NSF AI Institute for A. I. and Fundamental Interactions	Feb 2021
Seminar Series on String Phenomenology	Oct 2020

Contributed Workshop, Conference Talks, Posters

Parallel Session: Summer Workshop 2024 , The NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)	Aug 2024
String Data 2023 , Caltech	Dec 2023
Parallel Session: Summer Workshop 2023 , The NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)	Aug 2023
Poster Session: Theoretical Physics for Machine Learning , Aspen Center for Physics Feb 2023	
Poster session: Summer Workshop 2022 , The NSF IAIFI	Aug 2022
Parallel Session: String Phenomenology 2022 , University of Liverpool	Jul 2022

Contributed Seminars

Machine Learning Initiative Seminar , Perimeter Institute	Feb 2024
IPPP Seminar , Institute for Particle Physics Phenomenology, Durham University	Nov 2022
Oxford Dalitz Seminar in Fundamental Physics , U. Oxford	Nov 2022
Theoretical Particle Physics & Cosmology Seminar , King's College London	Oct 2022
Mathematics Seminar , City, University of London	Oct 2022
Theoretical Physics Seminar , Uppsala University	Oct 2022
Journal Club , The NSF AI Institute for A. I. and Fundamental Interactions	Sept 2022

Organization Workshop and Conferences

Generative AI for High & Low Energy Physics , Kavli Institute for Theoretical Physics, UC Santa Barbara, USA	Nov-Dec 2025
<i>- Member of the organizing committee and DEI coordinator</i>	

A Look into the Future through the Lens of AI Research, Perimeter Institute for Theoretical Physics, Canada
Apr 2025

- *Member of the principal organizing committee*

Future Horizons: Bridging AI, Quantum and New Materials, IVADO, Perimeter Institute, Institut Courtois and Mila Quebec, Canada
May 2024

- *Member of the organizing committee*

At the interface of physics, mathematics and artificial intelligence, Pollica Physics Center, Italy
May-June 2023

- *Member of the organizing committee*

Teaching Experience

TUTORIAL LEAD – Future Horizons: Bridging AI, Quantum and New Materials 2024

Machine Learning from a Physicist's Perspective: lectures by Anindita Maiti

TUTORIAL LEAD – The NSF IAIFI Summer School 2023

Normalizing Flows for Lattice Field Theory: lectures by Miranda Cheng

TEACHING ASSISTANT – Northeastern University, Boston, Massachusetts

PHYS 7325: Quantum Field theory 1 (Fall 2020, Fall 2019)

PHYS 5115: Quantum Mechanics (Spring 2020, Spring 2019)

PHYS 3601: Classical Dynamics (Fall 2018)

PHYS 2305: Thermo and Statistical Mechanics (Spring 2018)

PHYS 1155: Physics Lab for Engineering 2 (Fall 2017)

PHYS 3600: Advanced Physics Lab (multiple semesters)

Undergraduate Physics lab (multiple semesters)

TEACHING ASSISTANT – IIT Bombay, Mumbai, India

PH 117: Undergraduate Physics lab (Spring 2017)

EP 215: Undergraduate Electronics lab (Fall 2016)

Professional Service Activities

MEMBER, ANTI-RACISM WORKING GROUP & MENTAL HEALTH WORKING GROUP: Perimeter Institute for Theoretical Physics (Sept 2023 - Present).

ALUMNUS MENTOR: For undergraduate students in Engineering Physics major at IIT Bombay. (Sept 2022 - Dec 2024)

MEMBER, EARLY CAREER AND EQUITY COMMITTEE: The NSF AI Institute for Artificial Intelligence and Fundamental Interactions. (Jan 2021 - Dec 2022)

MEMBER, GRADUATE STUDENT COUNCIL: Northeastern University College of Science. (Sept 2020 - Aug 2022)

COORDINATOR & INITIATOR, [GRADUATE WOMEN IN PHYSICS SOCIETY](#) : Northeastern University Dept. of Physics. (Sept 2021 - May 2023)

REFeree: SynS & ML @ ICML2023; NeurIPS 2022 workshop on Machine Learning and the Physical Sciences; NeurIPS 2021 workshop on Machine Learning and the Physical Sciences; 'Foundations of Physics' Journal; NeurIPS 2020 workshop on Machine Learning and the Physical Sciences.

Summer Schools

- IAIFI Summer School, Aug 2022, The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

- Theoretical Advanced Study Institute in Particle Theory (TASI), June 2021, CU Boulder.

- Deep Learning Theory Summer School at Princeton, Jul 2021, Princeton University.

References

(1) PROF. ROGER MELKO, (*Email: rmelko@perimeterinstitute.ca*), Professor, Department of Physics & Astronomy, University of Waterloo, Associate Faculty, Perimeter Institute for Theoretical Physics, Canada Research Chair in Computational Quantum Many-Body Physics

(2) PROF. JAMES HALVERSON, (*Email: j.halverson@northeastern.edu*),
Associate Professor, Dept. of Physics, Northeastern University,
The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

(3) PROF. CENGIZ PEHLEVAN, (*Email: cpehlevan@seas.harvard.edu*),
Assistant Professor, Dept. of Applied Mathematics, Harvard University.
Kempner Institute Associate Faculty.

(4) PROF. MATTHEW SCHWARTZ, (*Email: schwartz@g.harvard.edu*),
Associate Professor, Dept. of Physics, Harvard University,
The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

(5) PROF. FABIAN RUEHLE, (*Email: f.ruehle@northeastern.edu*),
Assistant Professor, Dept. of Physics and Dept. of Mathematics, Northeastern University,
The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

(6) PROF. RO JEFFERSON, (*Email: r.jefferson@uu.nl*),
Assistant Professor, Institute for Theoretical Physics,
Department of Information and Computing Sciences, Utrecht University.

Technical skills

Programming languages: Python, C, C++, Mathematica, Matlab, Pytorch.